Tricorder Sensor Wiring - Nexys A7-100T (bench sheet)

Pin-exact vs hw/constraints/tricorder-nexys-a7-100t.xdc. All header logic 3.3 V. Pmod: pins 5,11=GND 6,12=VCC3V3.

Pmod JA - I2C bus + slow control

Pin	FPGA	Signal	Wire to
JA1	C17	i2c_scl	STEMMA QT chain SCL (yellow) + OV7670 SIOC
JA2	D18	i2c_sda	STEMMA QT chain SDA (blue) + OV7670 SIOD
JA3	E18	sipm_pulse	SiPM comparator out (async in)
JA4	G17	sipm_thresh_pwm	SiPM threshold RC filter
JA7	D17	afe_gain_pwm	AD8332 GAIN (via RC)
JA8	E17	afe_hilo	AD8332 HILO
JA9	F18	sonar_tx_a	piezo drive A (via driver)
JA10	G18	sonar_tx_b	piezo drive B (crosstalk swap target)

I2C daisy chain, 4.7k pull-ups at connector; breakouts have their own: BME688 0x76/77 -> SCD-41 0x62 -> VEML7700 0x10 -> AS7341 0x39 -> MAX30105 0x57. OV7670 SCCB 0x21 on same two wires (RST/PWDN strapped on module). No address conflicts.

Pmods JB + JC - OV7670 camera

Pin	FPGA	Signal	Pin	FPGA	Signal
JB1	D14	cam_xclk (24MHz out)	JB10	H16	cam_pclk ** MRCC, do not move
JB2	F16	cam_href	JC1	K1	cam_data[4]
JB3	G16	cam_vsync	JC2	F6	cam_data[5]
JB4	H14	cam_data[0]	JC3	J2	cam_data[6]
JB7	E16	cam_data[1]	JC4	G6	cam_data[7]
JB8	F13	cam_data[2]			
JB9	G13	cam_data[3]			

Keep cam_pclk on a 10 cm run. SCCB goes to JA1/JA2, not JB/JC.

Pmods JC + JD - AD9226 ADC / AD9708 DAC shared bus (DAC drives bits 7:0)

Pin	FPGA	Signal	Pin	FPGA	Signal
JD1	H4	adcdac_data[0]	JD9	G2	adcdac_data[6]
JD2	H1	adcdac_data[1]	JD10	F3	adcdac_data[7]
JD3	G1	adcdac_data[2]	JC7	E7	adcdac_data[8]
JD4	G3	adcdac_data[3]	JC8	J3	adcdac_data[9]
JD7	H2	adcdac_data[4]	JC9	J4	adcdac_data[10]
JD8	G4	adcdac_data[5]	JC10	E6	adcdac_data[11]

Clock: adcdac_clk on JXADC XA_N4 (A18), FPGA-sourced. DERATED to <= 40 MS/s behind 200-ohm Pmod series resistors. 10 cm harness mandatory. AD8332 analog out -> AD9226 in over SMA (RG316 + pigtails).

Pmod JXADC - PDM mic array + ADC clock (digital use, 100-ohm series)

Pin	FPGA	Signal	Wire to
XA_P1 /1	A13	pdm_clk	all mic CLK, fan out on breadboard
XA_N1 /7	A14	pdm_data[0]	mic 0 DATA
XA_P2 /2	A15	pdm_data[1]	mic 1 DATA
XA_N2 /8	A16	pdm_data[2]	mic 2 DATA
XA_P3 /3	B16	pdm_data[3]	mic 3 DATA
XA_N3 /9	B17	pdm_data[4]	mic 4 DATA
XA_P4 /4	B18	pdm_data[5]	mic 5 DATA ** paired w/ adcdac_clk - watch crosstalk
XA_N4 /10	A18	adcdac_clk	AD9226/AD9708 CLK

On-board ADMP421 is the 7th mic (J5/H5/F5) - no harness wiring.

Sonar / piezo chain

```
sonar_tx_a (JA9) --+
                   +-- driver -- piezo disc (2-pin screw term, twisted pair)
sonar_tx_b (JA10) -+               |
                                   +-- AD8332 AFE (SMA) -- AD9226 (JC/JD)
  rx alt: TLV3501 comparator on breadboard -> GPIO
afe_gain_pwm (JA7) -- RC -- AD8332 GAIN      afe_hilo (JA8) -- AD8332 HILO
```

On-board (no harness)

Sensor	Interface	Pins
ADT7420 temp	I2C1 @ 0x4B	C14/C15 (+irq)

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ADXL362 accel	SPI	E15/F14/F15/D15 (+irq)
ADMP421 PDM mic	PDM	J5/H5/F5
LAN8720A PHY	RMII	dedicated
S25FL128S flash	QSPI	dedicated

POWER - read before plugging

Everything on Pmod 3V3. SCD-41 has ~200 mA peaks: give it its own Pmod VCC pin (JA6 or JA12, alone) - do NOT stack with the camera. Piezo drive and SiPM bias stay OFF the Pmod rail. Header budget: 40/40 signal pins used - zero spares.

Tricorder Sensor Wiring - Nexys A7-100T (bench sheet)

Pin-exact vs hw/constraints/tricorder-nexys-a7-100t.xdc. All header logic 3.3 V. Pmod: pins 5,11=GND 6,12=VCC3V3.

Bring-up checklist (work in this order)

UART 115200 prints SENS: transitions on detection; UDP listener (port 5001) shows the key go live in ~2 s. No rebuilds for I2C steps.

0. Before anything

- [ ] Standalone boot streams telemetry (baseline sens_mask=16385, harness keys sentinels)
- [ ] Meter-check 3V3/GND at every Pmod plug BEFORE first insertion (5/11=GND, 6/12=VCC)
- [ ] Power off before plugging every harness change

1. Speaker (instant feedback)

- [ ] STEMMA speaker: Pmod 3V3+GND power, signal from AUD jack pigtail (tip=sig, sleeve=GND)
- [ ] Power-cycle: boot chirp audible; flip any switch: scan sound

2. I2C chain - ONE device at a time, power-cycle between additions

- [ ] VEML7700 first (full driver): light_lux live; cover it, value drops
- [ ] SCD-41 (own VCC dupont from JA12, chain cable data-only): co2_ppm after ~5 s; breathe, climbs
- [ ] BME688: UART PRESENT->ERROR = wiring proven (identity-only); bme_* read -2, NOT -1
- [ ] AS7341: same identity-only pattern
- [ ] MAX30105: same; i2c= count in telemetry = devices answering
- [ ] Full chain: 'I2C-EXT: walk -> 5 device(s)' on the 30 s UART walk

3. PDM mic array (SW0 -> PDM personality)

- [ ] Mic 0 only (clk + data[0]): mic_spl_db/energy live; snap fingers, energy jumps
- [ ] Add mics 1-5; verify each line by covering mics one at a time
- [ ] Watch pdm_data[5] during ADC runs (coupled w/ adcdac_clk); swap = sonar_tx_b/JA10

4. Camera (OV7670)

- [ ] SCCB only first (tap JA1/JA2): UART OV7670 PRESENT, cam_pid live
- [ ] Then JB/JC data harness, 10 cm on cam_pclk (CAM personality reserved; SCCB works today)

5. ADC/DAC bus (SW1 -> SONAR personality)

- [ ] AD9226 on JC/JD + adcdac_clk from JXADC10: adc_rate_ksps + min/max/rms live
- [ ] Grounded input: rms ~0, min/max tight; floating: noise
- [ ] adc_rail_cnt stays 0 with a real signal
- [ ] AD9708 shares bits 7:0 - only after ADC path proven

6. Sonar chain (last - most parts)

- [ ] Piezo TX pair on screw terminals -> JA9/JA10, twisted pair
- [ ] AD8332: HILO/GAIN from JA7/JA8, analog out -> AD9226 SMA
- [ ] SW1 fires 1 Hz bursts: sonar_bursts_cnt advances; book at ~30 cm: sonar_range_mm ~300
- [ ] SiPM eval last (own bias supply, SMA -> comparator -> JA3)

7. Regression after every session

- [ ] Switches off, power cycle: chirp, lease, 75-key frame, wired PRESENT, rest sentinels - never a fabricated in-range value